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Bachelor of Computer Applications

Semester – V

Business Analytics

Experiment No: 4

Title: Data Cleaning using Power Query

[GITHUB LINK](#)

Submitted By

Name: Adarsh Kasaudhan

Reg. No.:2411021240076

Date:09-09-2026

Submitted To:

Faculty Name: Mr. Aman Kumar Sharma

Department of Computer Application
Alliance University
Chandapura - Anekal Main Road, Anekal
Bengaluru – 562106

Experiment No.: 4

Data Cleaning using Power Query

Problem Statement

A cafe company has received customer transaction data containing missing values, blank cells, null values, and inconsistent entries. The objective is to clean and prepare the dataset using Power Query so that it becomes accurate, consistent, and ready for business analysis.

Methodology / Steps Followed

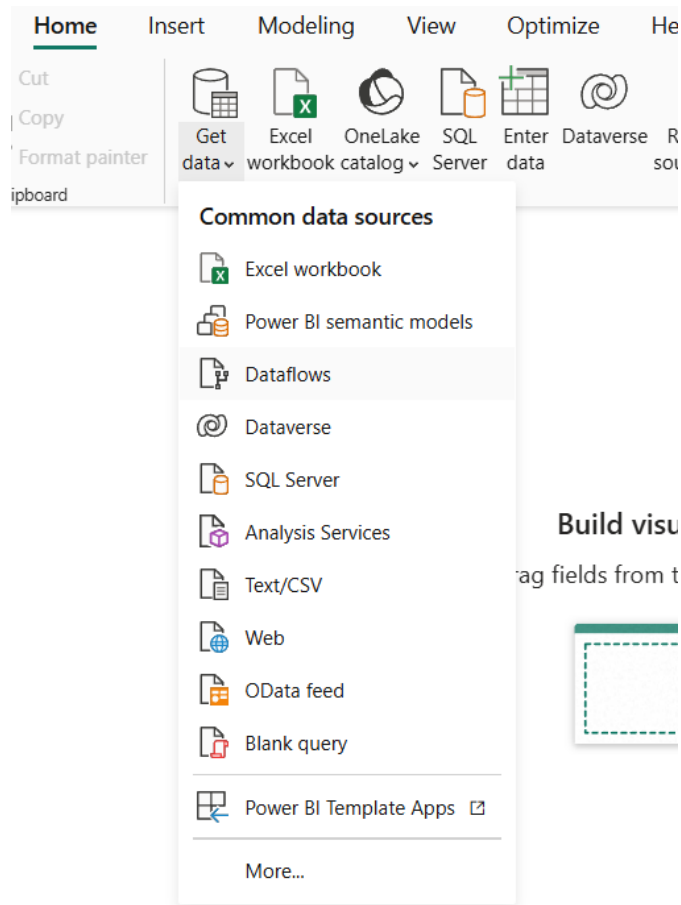
Step 1: Data Collection

- Collected the Dirty Cafe Sales dataset in CSV format.
- The dataset contains transaction details, product information, payment method, location, quantity, price, and transaction date.

Transaction ID	Item	Quantity	Price Per Unit	Total Spent	Payment Method	Location
TXN_1961373	Coffee	2	2	4	Credit Card	Tak
TXN_4977031	Cake	4	3	12	Cash	In-s
TXN_3160411	Coffee	2	2	4	Digital Wallet	In-s
TXN_2548360	Salad	5	5	25	Cash	Tak
TXN_7619095	Sandwich	2	4	8	Cash	In-s
TXN_2847255	Salad	3	5	15	Credit Card	In-s
TXN_6769710	Juice	2	3	6	Cash	In-s
TXN_3709394	Juice	4	3	12	Cash	Tak
TXN_3567645	Smoothie	4	4	16	Credit Card	Tak
TXN_5132361	Sandwich	3	4	12	Digital Wallet	Tak
TXN_9400181	Sandwich	5	4	20	Cash	In-s
TXN_5183041	Cookie	5	1	5	Credit Card	In-s
TXN_5695074	Juice	4	3	12	Credit Card	Tak
TXN_8853997	Smoothie	2	4	8	Digital Wallet	Tak
TXN_1080432	Salad	2	5	10	Credit Card	In-s
TXN_8078640	Juice	4	3	12	Digital Wallet	In-s
TXN_8614868	Smoothie	5	4	20	Digital Wallet	Tak
TXN_6421134	Sandwich	4	4	16	Cash	Tak
TXN_3068204	Cookie	1	1	1	Credit Card	Tak
TXN_2621580	Tea	2	1.5	3	Cash	In-s
TXN_4756776	Cake	2	2	4	Credit Card	In-s

Step 2: Data Import

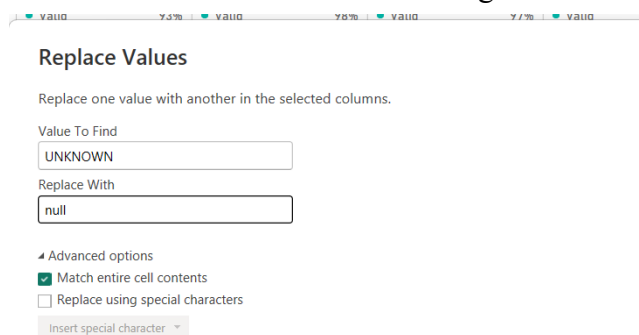
- Opened Microsoft Power BI Desktop.
- Selected Get Data → Text/CSV.
- Imported the dirty_cafe_sales.csv dataset.
- Clicked Transform Data to open Power Query Editor.



Step 3: Data Cleaning and Transformation

The dataset was cleaned using the following Power Query operations:

- Promoted the first row as column headers.
- Changed the data type of each column.
- Replaced invalid values (UNKNOWN and ERROR) with null.
- Filtered and removed empty, blank, and null records.
- Removed rows containing data errors.
- Verified the cleaned dataset before loading it into Power BI.



Source	⚙
✕ Changed Type	
Promoted Headers	⚙
Changed Type1	
Replaced Value	⚙
Replaced Value1	⚙
Filtered Rows	⚙
Changed Type2	
Removed Errors	
Filtered Rows1	⚙
Changed Type3	
Removed Errors1	
Filtered Rows2	⚙
Changed Type4	
Removed Errors2	
Filtered Rows3	⚙
Filtered Rows4	⚙
Filtered Rows5	⚙
Filtered Rows6	⚙
Replaced Value2	⚙
✕ Filtered Rows7	⚙

Step 4: Data Validation

After cleaning:

- Verified that all numeric columns were correctly formatted.
- Confirmed the transaction date was in Date format.
- Ensured no blank or invalid records remained.
- Loaded the cleaned dataset into Power BI for analysis.

Transaction ID	Item	Quantity	Price Per Unit	Total Spent	Payment Method	
TXN_1961373	Coffee	2	2	4	Credit Card	Tak
TXN_4977031	Cake	4	3	12	Cash	In-s
TXN_3160411	Coffee	2	2	4	Digital Wallet	In-s
TXN_2548360	Salad	5	5	25	Cash	Tak
TXN_7619095	Sandwich	2	4	8	Cash	In-s
TXN_2847255	Salad	3	5	15	Credit Card	In-s
TXN_6769710	Juice	2	3	6	Cash	In-s
TXN_3709394	Juice	4	3	12	Cash	Tak
TXN_3567645	Smoothie	4	4	16	Credit Card	Tak
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TXN_5183041	Cookie	5	1	5	Credit Card	In-s
TXN_5695074	Juice	4	3	12	Credit Card	Tak
TXN_8853997	Smoothie	2	4	8	Digital Wallet	Tak
TXN_1080432	Salad	2	5	10	Credit Card	In-s
TXN_8078640	Juice	4	3	12	Digital Wallet	In-s
TXN_8614868	Smoothie	5	4	20	Digital Wallet	Tak
TXN_6421134	Sandwich	4	4	16	Cash	Tak
TXN_3068204	Cookie	1	1	1	Credit Card	Tak
TXN_2621580	Tea	2	1.5	3	Cash	In-s
TXN_4776776	Cake	2	2	4	Credit Card	In-s

Output

The cleaned dataset provides:

- Accurate transaction records
- Standardized payment methods and locations
- Valid quantity and price values
- Correctly formatted dates
- Error-free data ready for visualization and analysis

Business Insights

After cleaning the dataset:

- Missing and null values were successfully removed.
- Invalid entries (UNKNOWN, ERROR) were corrected.
- Data types were standardized across all columns.
- The dataset became consistent and suitable for business reporting.
- Clean data improves the accuracy of dashboards and decision-making.

Conclusion

This experiment successfully cleaned and prepared the Dirty Cafe Sales dataset using Power Query. Missing values, blank cells, null records, and inconsistent entries were handled through data transformation techniques such as replacing values, filtering rows, removing errors, and correcting data types. The resulting dataset is reliable and ready for further analysis in Power BI.

Learning Outcomes

Through this experiment, I learned:

- Importing CSV datasets into Power BI
- Using Power Query for data cleaning
- Handling null, blank, and missing values
- Replacing inconsistent entries
- Removing errors and validating data quality
- Preparing datasets for accurate business analytics